

China Healthcare

BCI Expert Call Takeaways: Commercial Validation, Not Technology Demonstration, Is the Next Inflection Point

We hosted three brain-computer interface (BCI) expert call sessions in the last few days and the experts we invited came from leading BCI companies in China, hold key management positions and on average possess about 10 years of experience in this field. Our most important takeaway based on the expert speakers' views is that BCI is transitioning from technology validation to clinical and commercial validation, but that the pace of monetization is likely to be slower and more volatile than recent headline regulatory news would suggest. The BCI competitive landscape should not be viewed as a pure technology race in which more advanced technology - such as higher channel count or more sophisticated signal acquisition - automatically translates into greater commercial value. Rather, the industry is increasingly segmenting by patient severity, clinical need and device invasiveness. Non-invasive BCI is likely sufficient for the majority of lower-acuity rehabilitation and training applications; semi-invasive systems may occupy the most attractive risk-benefit position for patients with severe functional impairment; penetrating BCI will remain concentrated in a small population requiring the highest signal fidelity; and ultrasound is, for now, more realistically a neuromodulation platform than a read & write BCI. The experts tend to believe that the first approved invasive BCI products in China may serve primarily as policy and clinical infrastructure milestones rather than meaningful revenue generators. They think that the next stage of sector differentiation will be determined by whether companies can demonstrate incremental efficacy over standard of care (SoC) for various disease indications, accumulate scalable patient data for further development, secure government and commercial insurance reimbursement, and convert hospital/home access into repeatable utilization. Here are a few key takeaways from the calls.

- **The emerging clinical hierarchy will be determined by the trade-off between signal quality and invasiveness.** Non-invasive EEG (electroencephalogram)-based BCI has a fundamental signal-to-noise disadvantage, but for a large proportion of stroke patients with mild-to-moderate impairment, the expert feedback suggests that EEG may already be sufficient for rehabilitation, and subjecting these patients to neurosurgery would offer limited incremental value. Semi-invasive systems appear to occupy a potentially attractive middle ground: by avoiding penetration of the dura, they reduce surgical complexity and long-term biological risk while offering materially better signal quality than scalp EEG. This could make semi-invasive BCI the most commercially relevant implantable route for severe motor impairment, provided that it can demonstrate functional outcomes beyond those achievable through intensive rehabilitation activities plus non-invasive BCI. Penetrating systems such as Neuralink offer the highest signal resolution, but their addressable population is likely to remain narrow because the incremental functional benefit must be large enough to justify substantially greater surgical, ethical and long-term safety burdens.
- **Regulatory approval should be viewed as the beginning of product validation rather than the end of development.** The experience of the first approved semi-invasive system is particularly instructive: according to one expert, rapid approval carries substantial policy and symbolic value, yet commercial adoption may remain constrained by limited functional

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differentiation, uncertain physician conviction, and a substantial patient self-pay burden. This suggests a meaningful disconnect between regulatory progress and revenue realization. More BCI products could reach the market around 2027, but early commercial adoption is still likely to resemble controlled clinical deployment or advanced research rather than conventional high-volume medical-device adoption. According to the expert, a genuine commercial inflection point would require a reinforcing cycle in which post-launch clinical use generates broader datasets, better algorithms and more stable outcomes, which in turn support indication expansion, physician adoption and reimbursement.

- **The technical bottleneck is shifting from individual hardware components towards system-level integration and longitudinal signal robustness.** The current development cycle has been enabled by advances in semiconductor miniaturization, packaging, high-density feedthroughs and electrode fabrication, but the expert feedback indicates that many BCI systems remain a collection of independently optimized modules rather than a fully integrated product. This is an important distinction: a high-channel-count electrode has limited value if the implant, acquisition chip, wireless transmission, decoding algorithm and effector cannot operate reliably as one system over years. The same logic applies to headline channel count. For motor-function restoration, the experts suggest that roughly 200 channels may already be clinically useful and that approximately 512 channels could address the large majority of current requirements. This implies diminishing returns from pursuing channel count alone. The more defensible competitive advantages may instead lie in chronic signal stability, implant durability, low-power processing, reproducible surgical implantation and closed-loop system performance.
- **AI can create real value in BCI, but its near-term impact will be highly asymmetric across technology routes and might be overstated when described as a universal solution.** The non-invasive BCI expert argued that large AI models can materially shorten algorithm-development cycles and accelerate individual calibration, while the invasive BCI expert remained skeptical that existing large-model architectures can solve the fundamental instability of continuous, patient-specific neural signals. In non-invasive BCI, where products often solve more constrained classification or personalization problems and suffer from noisy signals, AI can improve denoising, reduce calibration requirements and make existing hardware materially easier to use. In implanted/invasive BCI, however, the core bottleneck is not simply AI model sophistication: neural signals differ between individuals, change with patient state and may drift over time after implantation. A generalizable “brain foundation model” therefore requires large, standardized and longitudinal multimodal datasets that do not yet exist. Plus, what algorithm will be best for a BCI-related model is also unknown. According to the expert, near-term value of AI for BCI should accrue to companies using AI to reduce patient-specific training and improve real-world robustness, while the much larger promise of universal neural decoding or even “brain model” should be viewed as a longer-term thesis.
- **The domestic supply chain opportunity is rising and might be important to support long-term needs.** For BCI components, China needs to catch up in analog front-end chips for EEG, implantable acquisition chips and certain highly specialized biocompatible materials, while its strengths are more visible in non-invasive electrode manufacturing. However, the commercial significance of each localization opportunity differs sharply. Implant-grade polyimide and biocompatible silicone may be strategically important, but current industry demand is too small to support a large standalone upstream market. The more attractive opportunities are likely to emerge where BCI requirements overlap with larger markets, such as low-noise analog front ends, flexible electronics, high-density packaging, MEMS (micro-electromechanical systems) transducers and low-power signal processing. Our experts also expect local production to accelerate for strategic and

data-security reasons, even where domestic components initially trail imported products in absolute performance.

- **Ultrasound represents the highest long-term technical optionality among emerging routes.** The ultrasound session presented an ambitious vision of broader brain coverage with ultrasound technology, multimodal foundation models and eventual whole-brain read-write capability, but the current commercial pathway is much more concrete on the “write” side than the “read” side. Low-intensity focused ultrasound (LIFU) can modulate specific brain regions non-invasively, creating real applications in chronic pain management, psychiatric disease control, stroke rehabilitation and blood-brain-barrier opening. In contrast, high-resolution transcranial reading remains constrained by the skull, and blood flow-based signals introduce a temporal gap relative to direct electrical activity. Chronic pain could be a sensible first indication for ultrasound-based BCI, and its success will depend on conventional endpoints including clinical efficacy, durability, and safety. The expert is optimistic about this indication as the target population is large and there is a clear unmet need for non-opioid alternatives.

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